G1020 sanity check: in-domain performance of five pre-trained CNNs

Train / validation / test = 714 / 153 / 153 images (stratified 70/15/15; test set has 44 glaucoma images). Seeds 42, 7, 123. Image size 224, batch 32, up to 15 epochs, learning rate 0.0001. Sensitivity, specificity and accuracy use the Youden threshold chosen on the validation set. Values are mean ± sample SD over the three seeds.

Table A. Mean ± SD over 3 seeds

Model	Val AUC	Test AUC	Sensitivity	Specificity	Accuracy
ResNet50	0.571 ± 0.034	0.511 ± 0.066	0.386 ± 0.138	0.651 ± 0.106	0.575 ± 0.053
DenseNet121	0.603 ± 0.040	0.490 ± 0.036	0.432 ± 0.364	0.547 ± 0.295	0.514 ± 0.106
MobileNetV2	0.575 ± 0.047	0.529 ± 0.092	0.303 ± 0.129	0.722 ± 0.190	0.601 ± 0.108
EfficientNet-B0	0.554 ± 0.051	0.523 ± 0.068	0.750 ± 0.295	0.275 ± 0.313	0.412 ± 0.140
InceptionV3	0.605 ± 0.069	0.563 ± 0.029	0.492 ± 0.265	0.554 ± 0.241	0.536 ± 0.103

Table B. Test AUC per seed

Model	Seed 42	Seed 7	Seed 123
ResNet50	0.587	0.473	0.473
DenseNet121	0.531	0.467	0.471
MobileNetV2	0.617	0.433	0.536
EfficientNet-B0	0.576	0.546	0.446
InceptionV3	0.555	0.539	0.596

Figure. ROC curves and AUC bars

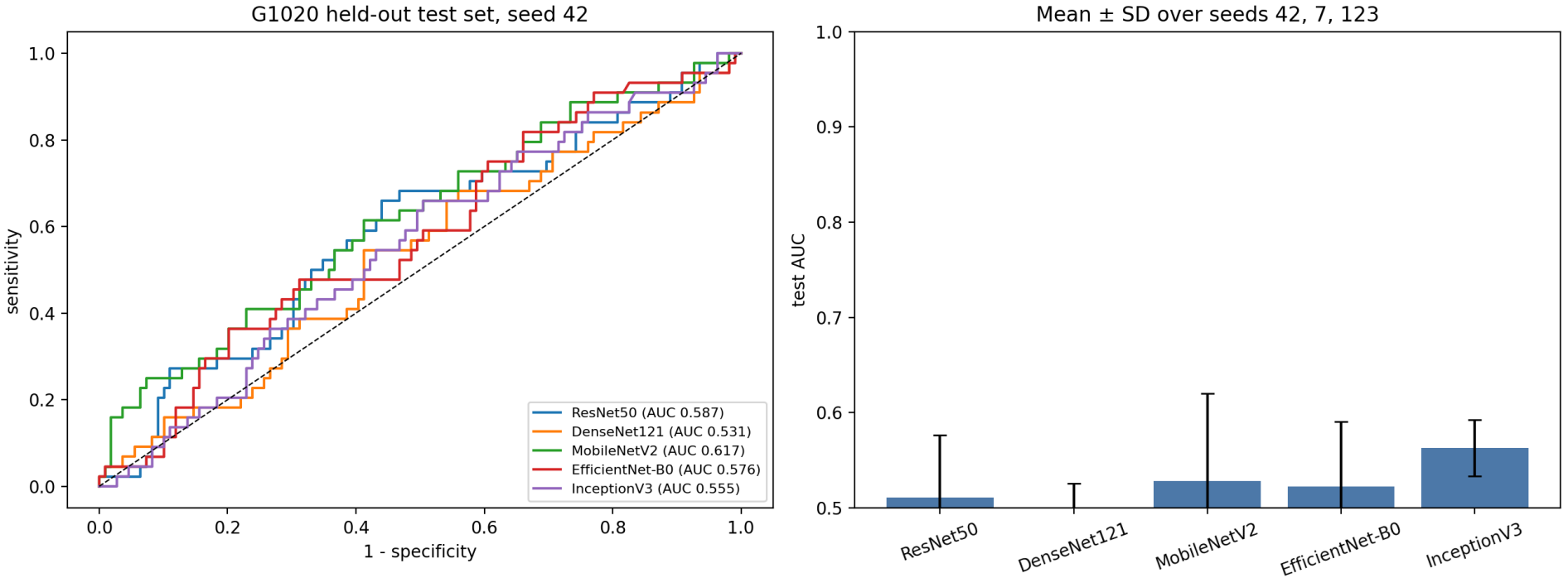


Table C. All 15 runs

Model	Seed	Test n	Test glauc.	Val AUC	Threshold	Test AUC	Sens	Spec	Acc
ResNet50	7	153	44	0.594	0.498	0.473	0.477	0.532	0.516
ResNet50	42	153	44	0.532	0.535	0.587	0.455	0.688	0.621
ResNet50	123	153	44	0.588	0.534	0.473	0.227	0.734	0.588
DenseNet121	7	153	44	0.635	0.63	0.467	0.068	0.826	0.608
DenseNet121	42	153	44	0.558	0.414	0.531	0.795	0.239	0.399
DenseNet121	123	153	44	0.614	0.38	0.471	0.432	0.578	0.536
MobileNetV2	7	153	44	0.627	0.545	0.433	0.341	0.55	0.49

Model	Seed	Test n	Test glauc.	Val AUC	Threshold	Test AUC	Sens	Spec	Acc
MobileNetV2	42	153	44	0.563	0.454	0.617	0.409	0.688	0.608
MobileNetV2	123	153	44	0.535	0.645	0.536	0.159	0.927	0.706
EfficientNet-B0	7	153	44	0.597	0.496	0.546	0.409	0.633	0.569
EfficientNet-B0	42	153	44	0.568	0.456	0.576	0.932	0.138	0.366
EfficientNet-B0	123	153	44	0.497	0.21	0.446	0.909	0.055	0.301
InceptionV3	7	153	44	0.654	0.354	0.539	0.727	0.294	0.418
InceptionV3	42	153	44	0.526	0.744	0.555	0.205	0.771	0.608
InceptionV3	123	153	44	0.635	0.279	0.596	0.545	0.596	0.582